

# Homework 2: Image segmentation without user input

December 16, 2025

## 1 Exemplary Visualization of Grad-CAM Output

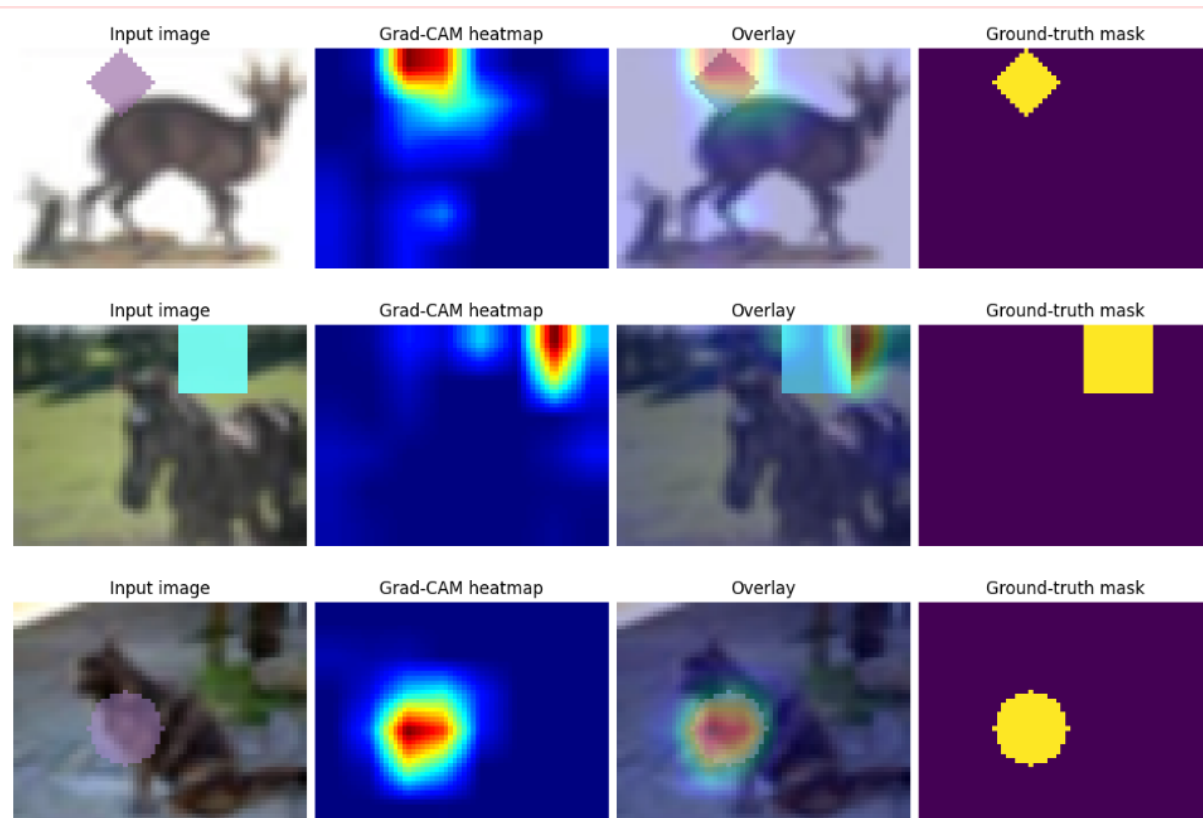


Figure 1: Grad-CAM heatmaps overlaid on input images.

## 2 Pipeline Approaches

Two distinct pipelines were evaluated to generate prompts for the Segment Anything Model (SAM) based on Grad-CAM activations:

### 1. **FSamPipeline (Foreground-only):**

Here I normalized the map to values  $[0,1]$  and binarized it. Every value above 0.6 got the label true and the rest false, to isolate the most important pixels. I got the idea from the paper where thresholding was used in weakly-supervised localization (4.1 Weakly-supervised Localization). The geometric centroid of this region is calculated and used as the foreground prompt for the segmentation model.

### 2. **FBSamPipeline (Foreground + Background):**

In addition to the foreground point identified by the previous algorithm, this approach selects a

negative prompt to suppress background noise. The image corner furthest from the foreground centroid was chosen, providing a reliable background reference without explicitly identifying the foreground box.

### 3 Metrics Presentation

Table 1: Performance comparison between FSamPipeline and FBSamPipeline.

| Metric                      | FSamPipeline | FBSamPipeline |
|-----------------------------|--------------|---------------|
| Avg Hit Rate                | 0.768        | 0.768         |
| Avg Distance                | 5.269        | 5.269         |
| Avg Intersection over Union | 0.802        | 0.795         |

### 4 Discussion and Potential Improvements

Just adding the furthest corner actually slightly decreased the average IoU. So, this is the first thing I would start improving. My next idea was to make some kind of box around my foreground point and think about some heuristic to identify the best point for the background part of the task. Maybe the threshold idea I got from the paper also could be investigated and improved; maybe different values would work better, or maybe a completely different idea should be implemented. I actually tested some different values, but my testing wasn't very extensive, so there is for sure some room for improvement.